

AI CAPSTONE PROJECT REPORT

AI Career Guidance Chatbot for Engineering Students

Submitted By:
Sarvesh Pednekar

Lenovo AI Internship Program

AI Career Guidance Chatbot for Engineering Students

Submitted By :

Name: Sarvesh Pednekar

Project Title :

AI Career Guidance Chatbot for Engineering Students

SDG Selected :

SDG 4 – Quality Education

Reason for Selection :

Quality education plays an important role in shaping students' careers. Many engineering students are confused about career paths, internships, higher studies, government jobs, and required skills. This project aims to provide personalized career guidance and educational support through an AI-powered chatbot.

1. Problem Statement :

Many students are uncertain about their future career paths and often lack proper guidance. They are unaware of suitable job opportunities, required technical skills, higher education options, internships, and entrepreneurship opportunities related to their engineering branch.

This problem exists among engineering students, diploma students, and fresh graduates. Without proper guidance, students may choose unsuitable career paths, miss valuable opportunities, and face difficulties in career planning.

2. Project Objective :

The objective of this project is to develop an AI-based Career Guidance Chatbot that helps students:

- Explore career opportunities based on their engineering branch
 - Identify important technical skills
 - Discover internship opportunities
 - Learn about higher studies options
 - Explore government job opportunities
 - Understand entrepreneurship opportunities
-

3. Proposed Solution :

The proposed solution is an AI Career Guidance Chatbot developed using Botpress.

The chatbot interacts with users through a simple conversation flow. Students select their engineering branch and area of interest. Based on the selected options, the chatbot recommends suitable career paths, skills, internships, higher studies options, government job opportunities, and entrepreneurship guidance.

The chatbot provides quick and structured guidance, helping students make informed career decisions.

4. How the System Works :

Step 1: User starts the chatbot.

Step 2: User selects their engineering branch.

Step 3: User selects their area of interest.

Step 4: The chatbot analyzes the selected options.

Step 5: Relevant career opportunities and skills are displayed.

Step 6: User selects future goals such as Placement, Higher Studies, Government Job, or Entrepreneurship.

Step 7: The chatbot provides detailed recommendations and guidance.

5. Project Features :

Branch-Based Guidance

Provides recommendations based on the student's engineering branch.

Career Recommendations

Suggests suitable career paths according to user interests.

Skill Recommendations

Displays important technical and professional skills required for selected careers.

Internship Guidance

Recommends internship platforms and learning opportunities.

Higher Studies Guidance

Provides information regarding higher education options.

Government Job Guidance

Suggests competitive exams and government career opportunities.

Entrepreneurship Guidance

Provides startup and business-related guidance for interested students.

Interactive Chat Interface

Simple and user-friendly conversational interface.

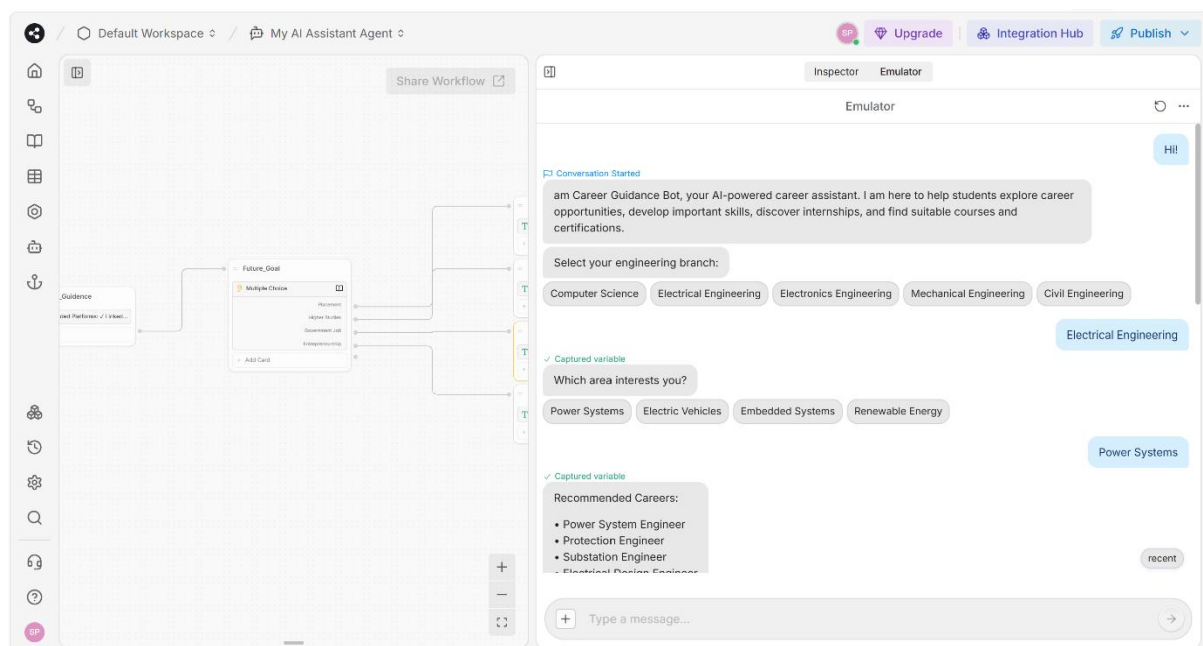
6. Technology Stack :

- Botpress
- Artificial Intelligence (AI)
- Natural Language Processing (NLP)
- Prompt Engineering
- Workflow-Based Chatbot Design

7. Users of the System :

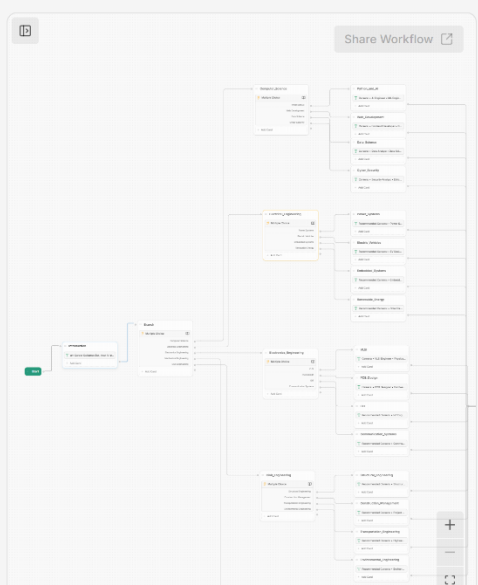
- Engineering Students
- Diploma Students
- Fresh Graduates
- Career Seekers

8. Project Screenshots :



Default WorkspaceMy AI Assistant Agent

Share Workflow



InspectorEmulator

Emulator

Captured variable

Recommended Careers:

- Power System Engineer
- Protection Engineer
- Substation Engineer
- Electrical Design Engineer

Skills:

- ETAP
- MATLAB
- Relay Protection
- Load Flow Analysis

Recommended Platforms:

- LinkedIn
- Internshala
- AICTE Internships
- Forage
- SkillDzire

Pick one or many from the list below

PlacementHigher StudiesGovernment JobEntrepreneurship

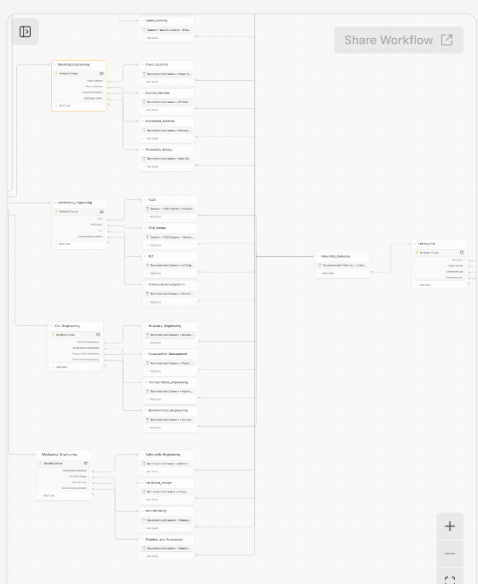
Government Job

Captured variable

Type a message...

Default WorkspaceMy AI Assistant Agent

Share Workflow



InspectorEmulator

Emulator

Captured variable

Recommended Opportunities:

- GATE for PSU Recruitment
- SSC JE
- RRB JE
- State Government Exams
- UPSC Engineering Services

Preparation:

- Technical Subjects
- Aptitude
- General Awareness

Thank you for using AI Career Guidance Bot.

Based on your choices, suitable career paths, skills, internships, and future opportunities have been recommended.

Best wishes for your career journey!

Conversation Ended

Captured variable

Type a message...

My AI Assistant Agent

Share Workflow

Workflow Diagram

Inspector

Emulator

Emulator

Pick one or many from the list below

PlacementHigher StudiesGovernment JobEntrepreneurship

Government Job

✓ Captured variable
go Transitioned - Main:undefined

Recommended Opportunities:

- ✓ GATE for PSU Recruitment
- ✓ SSC JE
- ✓ RRB JE
- ✓ State Government Exams
- ✓ UPSC Engineering Services

Preparation:

- Technical Subjects
- Aptitude
- General Awareness

go Transitioned - Main:undefined

Thank you for using AI Career Guidance Bot.
Based on your choices, suitable career paths, skills, internships, and future opportunities have been recommended.
Best wishes for your career journey!

Conversation Ended

Type a message...

My AI Assistant Agent

Share Workflow

Workflow Diagram

Inspector

Emulator

Emulator

Thank you for using AI Career Guidance Bot.
Based on your choices, suitable career paths, skills, internships, and future opportunities have been recommended.
Best wishes for your career journey!

Would you like another recommendation?

Start Again

✓ Captured variable
go Transitioned - Main:Introduction

Hello! 🤖
I am Career Guidance Bot, your AI-powered career assistant.
I am here to help students explore career opportunities, develop important skills, discover internships, and find suitable courses and certifications.
Please select your engineering branch to begin.

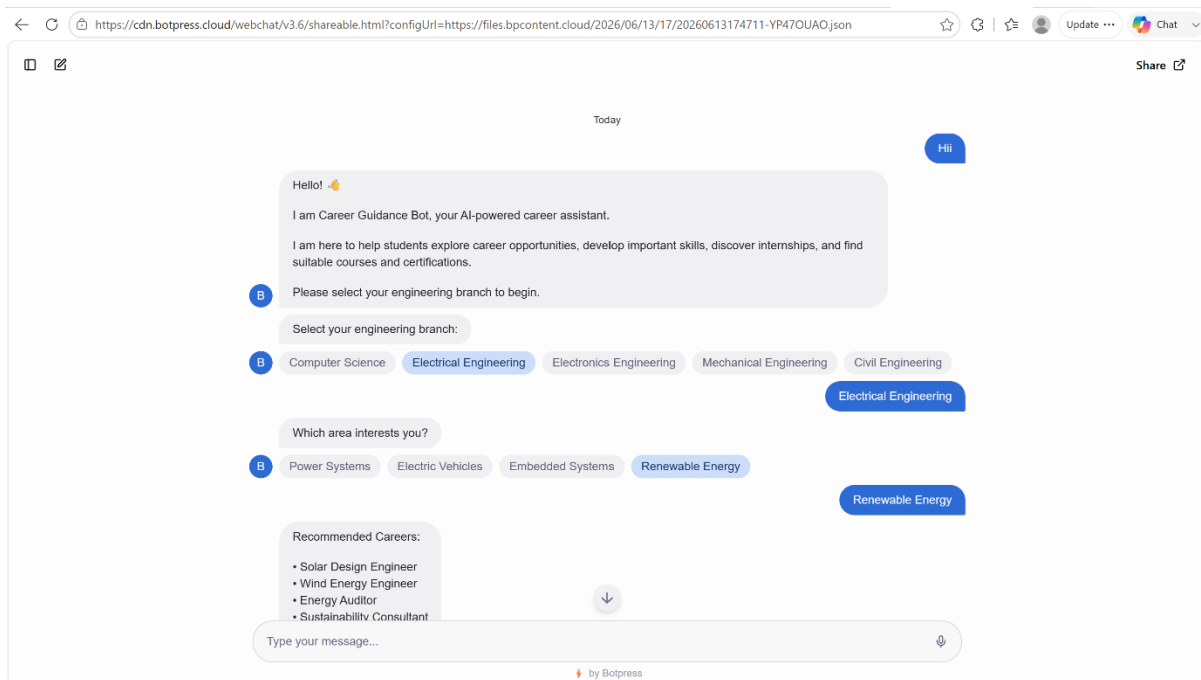
go Transitioned - Main:undefined
Capturing Variable - prompt - active

Select your engineering branch:

Computer ScienceElectrical EngineeringElectronics EngineeringMechanical EngineeringCivil Engineering

Waiting for User Input

Type a message...



9. Open Source Repository :

GitHub Repository Link:

<https://github.com/SarveshPednekar1025/AI-Career-Guidance-Chatbot.git>

10. Future Scope :

- Integration with real-time job portals
 - AI-powered resume analysis
 - Personalized career roadmap generation
 - Voice-based interaction
 - Mobile application development
 - Multi-language support
 - Real-time internship recommendations
 - Advanced AI career prediction models
-

11. Conclusion :

The AI Career Guidance Chatbot successfully assists students in exploring career opportunities, developing relevant skills, finding internships, and planning their future goals. The project supports SDG 4 (Quality Education) by providing accessible career guidance and helping students make informed decisions regarding their professional growth.